Faizan Naseem

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.00
Sliet Longowal	Class XII (Sant Longowal Institute of Engineering And technology)	2021	8.26%
UMS Machhahi	Class X (BSEB)	2018	70.80%

EXPERIENCE

Seimens Healthneeirs

- **Selected to represent Siemens Healthineers at FOMO27 (MICCAI 2027)**, a Foundation Model Challenge for Brain MRI, contributing to the development and evaluation of **robust, generalizable foundation models for brain MRI analysis** across diverse clinical settings.
- Contributed to the development of **3D brain MRI foundation-model approaches**, with a focus on leveraging **2D pretrained foundation models for 3D volumetric medical imaging without modifying the pretrained model weights**.
- Worked on adapting the pretrained representations and architecture to effectively process **volumetric brain MRI data**, targeting downstream tasks such as **3D anatomical segmentation** and improved generalization across clinical imaging domains.

Esparkbiz Pvt Ltd

- Developed and maintained responsive web applications using **React.js, Node.js, and Express.js**, with a focus on scalable and user-friendly interfaces.
- Designed and implemented **RESTful APIs** and integrated backend services with **MongoDB** for efficient data storage and retrieval.
- Implemented secure **authentication and authorization using JWT**, along with robust API validation and error-handling mechanisms.
- Collaborated with the development team to **debug, optimize, and deploy web applications** while adhering to software development best practices.

PROJECTS

Ai Based Automated Quantification of Epicardial FAT for Cardiac Health Monitoring

- Developing an AI-driven framework for automated quantification of Epicardial Adipose Tissue (EAT) from cardiac CT and MRI images using advanced deep learning and medical image analysis techniques.
- Awarded the prestigious IndiaAI Fellowship Grant worth INR 2,00,000 for the development of this ongoing AI-based healthcare research project.
- Implementing modality-specific preprocessing and semantic segmentation techniques to extract clinically relevant cardiac biomarkers with improved precision and reproducibility.
- Aiming to develop a fully automated, scalable, and operator-independent framework suitable for integration into real-world clinical workflows

Pneumonia Classification Using Deep Learning

- Designed and implemented a Convolutional Neural Network (CNN) for pneumonia detection using chest X-ray images.
- Trained the model using Stochastic Gradient Descent (SGD) with momentum for stable convergence and improved generalization.
- Applied image preprocessing and data augmentation techniques to enhance data quality and reduce overfitting.
- Utilized early stopping and learning rate scheduling to optimize training and improve validation performance.

Test-Time Adaptation of Vision Models

- Developed a **Test-Time Adaptation (TTA) framework** for vision models to improve performance under distribution shifts without requiring access to source-domain data.
- Implemented **entropy minimization-based adaptation** to enable models to learn from unlabeled test data during inference.
- Designed efficient adaptation strategies by updating **normalization statistics and channel-wise affine parameters** while preserving the model's core representations.
- Evaluated the approach under **real-world domain shifts and corrupted inputs**, demonstrating improved model robustness and generalization during inference.

SCHOLASTIC ACHIEVEMENTS

- **India AI Fellowship Grant (2,00,000)** for the AI-based Epicardial fat Quantification Project
- Qualified **GATE 2025** in Computer Science & Information Technology (CS)
- Qualified **GATE 2025** in Data Science & Artificial Intelligence (DA)

SKILLS

- **Languages & AI/ML:** Python, Java, JavaScript, SQL; PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, CNNs, Transformers, Foundation Models.
- **Medical AI & Computer Vision:** MONAI, nnU-Net, MED-SAM, MED-SAM2, DINOv2/DINOv3, Medical Image Segmentation, MRI/CT Analysis, Image Processing
- .**Web & Databases:** React.js, Node.js, Express.js, MongoDB, MySQL, REST APIs, JWT.
- **Tools & Infrastructure:** Git, GitHub, Docker, Kubernetes, CUDA, Linux, 3D Slicer.

RELEVANT COURSES

Deep Learning , Computer Vision, Biomedical Imaging, Sensors and Transducer in Healthcare, Magnetic Resonance Imaging (MRI) , Accident and Emergency Informatics

POSITIONS OF RESPONSIBILITY

- OCS placement Volunteer 2026-2027
- Operations Coordinator - 58th Inter IIT Sports Meet Aug 2025 - Feb 2026